

What should you know before buying a projector today?

- **Define the purpose.** Start by understanding your needs. If you're seeking a true TV alternative, focus on ultra short throw laser TVs, not just traditional projectors.
- **Know the global shift.** In the West and parts of Asia, 250cm to 300cm (100-120 inch) TVs are used mainly for commercial displays. For homes, laser TVs are gaining popularity due to their superior visuals, enhanced eye comfort, and portability.
- **Growing awareness in India.** Laser TVs entered India around 2019, and interest is rising. Consumers are researching more, and demand is growing—especially for compact, smart, large-screen alternatives.
- **The experience gap.** Many regret buying large LCD/LED TVs after experiencing modern laser projectors. The jump in size, comfort, and built-in sound is convincing more users to switch.
- **What's ahead.** While current models don't replace all TVs, the next wave—due within a year—could make screenless TVs a serious choice for many households.

Video, Disney + , and more.

Today's smart projectors are not just about image, they are about convenience, intelligence, and seamless integration into digital lifestyles.

Can projectors replace large TVs?

Projectors have long taken a back seat to TVs in home entertainment, but that is beginning to change, particularly for those wanting screens larger than 2.5-metre (100-inches). While 140cm to 190cm (55- to 75-inch) TVs remain popular, an increasing number of consumers are opting for projectors to enjoy an immersive, cinema-style experience.

However, not all projectors can replace a television. For a 140cm (75-inch) display, a TV may still be the more practical choice. However, for screen sizes above 250cm (100-inches), modern projectors—especially ultra-short-throw models—offer distinct advantages. Brands like Dangbei, Valerian, JMGO, and Hisense now offer laser projectors that produce bright, high-resolution images directly onto walls or screens. While they can project onto bare walls, pairing them with a quality screen enhances performance, rivaling premium TVs from LG, Samsung, or Sony.

Comfort is another key benefit. Large TVs emit heat and direct light, which may strain the eyes. Projectors reflect light, creating a softer, more eye-friendly experience. They

are also more portable, a 250cm (100-inch) TV is bulky and hard to move, whereas a projector can be unplugged and carried like a large laptop.

Image quality, however, depends heavily on the screen used. Standard matte white screens, commonly found in offices, often fail to showcase the capabilities of modern laser projectors. Fresnel screens, built with layered optics that include a Fresnel lens, enhance clarity and brightness. The result can resemble a backlit TV, with some users describing it as a complete visual upgrade.

While projectors are unlikely to replace smaller TVs anytime soon, for large-format home viewing, compact laser projectors combined with high-quality screens now present a compelling, and often superior, alternative.

What comes next?

Smart features have become standard in today's laser projectors. Built-in platforms like Google TV, AI enhancements, filmmaker modes, and premium audio are now expected. While current models are already advanced, there is still scope for improvement in size, portability, and daily usability.

In India, however, awareness remains relatively low. Many still associate projectors with bulky, outdated models used in classrooms or conference rooms. In reality, today's home projectors are

compact, intelligent, and capable of replacing large TVs. The leap in brightness, image clarity, and usability compared to older models is significant.

Audio quality has also improved. Early smart projectors featured Harman speakers, later replaced by Denon and Dynavox. The next generation will include integrated speakers from Divya Lane, a European brand. Many current models already support Dolby Atmos for virtual surround sound. Upcoming systems will support single-connection speaker setups, eliminating the need for a separate AV receiver and simplifying installation.

Visually, modern laser projectors significantly outperform older models. While core mechanics have not changed dramatically, user experience continues to improve through innovation. These devices are no longer just display tools—they are becoming all-in-one entertainment hubs for the home.

Projectors have evolved far beyond their traditional role. They have emerged as powerful, flexible, and intelligent alternatives to large-screen televisions. Powered by technologies such as ALPD, Trichroma, and MALC, today's laser projectors deliver cinema-grade visuals with enhanced portability, integrated smart features, and AI-driven ease of use. Whether it is a compact pico projector for mobile streaming or a short-throw model for immersive home viewing, the move toward screenless televisions is well underway. As awareness grows and adoption spreads, projectors are set to become not just replacements for big TVs but a redefinition of how we experience visual content at home. **EFY**

This article is based on a conversation with Sushil Motwani, founder of Aytexcel Pvt Ltd and official India representative for Formovie projectors, who has helped bring projection technology to users in India. Co-author, Nidhi Agarwal is Senior Technology Journalist at EFY with a deep interest in embedded systems, development boards, and IoT cloud solutions.